Finance Support Triage Agent

Course Name: Agentic AI

Institution Name: Medicaps University – Datagami Skill Based Course

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1. Problem Statement s Objectives

Problem Statement

In the financial services domain, organizations handle a vast number of customer queries and support requests through multiple communication channels such as emails, chat systems, and APIs. These queries typically relate to transactions, account management, fraud detection, billing issues, and technical assistance. Managing such a high volume of requests efficiently and accurately has become a significant challenge.

Most existing systems rely heavily on manual processing or basic rule-based automation. These approaches are often inadequate in handling the complexity and variability of natural language queries. As a result, several issues arise, including delayed response times, inconsistent classification of queries, improper routing to relevant departments, and failure to identify high-priority or urgent cases.

Furthermore, the lack of contextual understanding in traditional systems leads to generic or incorrect responses, reducing the overall quality of customer support. This not only affects customer satisfaction but also increases operational costs due to repeated interactions and inefficient use of human resources. Critical financial concerns, such as fraud alerts or transaction failures, may also be overlooked or delayed, potentially leading to serious consequences.

Therefore, there is a strong need for an intelligent and automated solution that can efficiently process, classify, and respond to customer queries. Such a system should be capable of understanding user intent, extracting relevant information, determining urgency, and either generating accurate responses or

escalating issues appropriately.

The development of an AI-powered Finance Support Triage Agent aims to address these challenges by leveraging advanced technologies such as Natural Language Processing (NLP), Named Entity Recognition (NER), and Large Language Models (LLMs). This system will enhance the efficiency, accuracy, and responsiveness of financial customer support services.

Project Objectives

The primary objective of this project is to design and develop an intelligent **Finance Support Triage Agent** that automates the processing and management of customer support queries in the financial domain. The system aims to enhance efficiency, accuracy, and response time while reducing manual intervention.

The specific objectives of the project are as follows:

- To **automatically classify incoming customer queries** based on their intent (e.g., transaction issue, account support, fraud alert, billing query) using Natural Language Processing techniques.
- To **identify and prioritize queries based on urgency**, ensuring that critical issues such as fraud detection or failed transactions are handled promptly.
- To **extract relevant information** such as account details, transaction IDs, and dates using Named Entity Recognition

- (NER) for better context understanding.
- To **generate accurate and context-aware responses** using Large Language Models (LLMs), improving the quality of customer interaction.
- To **automate workflow orchestration**, including routing queries to appropriate departments or triggering escalation mechanisms when necessary.
- To **reduce response time and operational costs** by minimizing manual effort and improving system efficiency.
- To **ensure scalability and adaptability** of the system so that it can handle increasing volumes of queries and integrate with multiple communication channels.
- To **improve overall customer satisfaction** by providing faster, more accurate, and consistent support services.

Scope of the Project

The scope of this project focuses on the design and development of an AI-powered **Finance Support Triage Agent** that automates the handling of customer queries within the financial services domain. The system is intended to improve efficiency, accuracy, and response time by leveraging advanced technologies such as Natural Language Processing (NLP), Named Entity Recognition (NER), and Large Language Models (LLMs).

The project includes the following key aspects:

- The system will **process incoming customer queries** from multiple sources such as email, chat interfaces, or APIs and convert them into a structured format for further analysis.
- It will **classify queries based on intent**, such as transaction issues, account-related problems, fraud alerts, and general inquiries.
- The solution will **identify the urgency level** of each query and prioritize critical cases to ensure timely resolution.
- It will **extract important entities** like account numbers, transaction IDs, dates, and amounts to build contextual understanding.
- The system will **generate appropriate responses** automatically or assist support agents with suggested replies using AI models.
- It will **route queries to relevant departments** or trigger escalation workflows when human intervention is required.

1. Proposed Solution

To address the challenges associated with manual and inefficient handling of financial customer support queries, this project proposes the development of an intelligent **AI-powered Finance Support Triage Agent**. The system is designed to automate the end-to-end process of receiving, analyzing, classifying, and responding to customer queries with minimal human intervention.

The proposed solution leverages advanced technologies such as **Natural Language Processing (NLP)**, **Named Entity Recognition (NER)**, and **Large Language Models (LLMs)** to understand user queries and generate context-aware responses.

The overall working of the system is described as follows:

- The system will first **collect incoming queries** from multiple channels such as email, chat, or APIs and preprocess the data to remove noise and standardize the input.
- A **classification engine** will analyze the query to determine the user's intent (e.g., transaction issue, account problem, fraud alert) and categorize it accordingly.
- The system will then perform **urgency detection** to prioritize critical queries that require immediate attention.

- Using **Named Entity Recognition (NER)**, the system will extract key information such as account numbers, transaction IDs, dates, and monetary values to build contextual understanding.
- A **context builder module** will combine extracted entities with historical or session data to create a complete representation of the query.
- Based on this context, a **Large Language Model (LLM)** will generate an accurate and relevant response or suggest possible solutions.
- A **workflow orchestrator** will decide the next action, such as:
    - Sending an automated reply
    - Routing the query to the appropriate department
    - Escalating high-priority issues to human agents
- Finally, the system will **deliver the response** through the appropriate communication channel and log the interaction for future reference and analysis.

Key Features

The proposed Finance Support Triage Agent includes several intelligent and automation-driven features designed to improve the efficiency and effectiveness of customer support operations in the financial domain. The key features of the system are as follows:

- **Automated Query Classification**
  The system automatically categorizes incoming customer queries based on their intent, such as transaction issues, account-related problems, fraud alerts, or general inquiries, using Natural Language Processing techniques.
- **Urgency Detection and Prioritization**
  It identifies the urgency level of each query and prioritizes critical cases to ensure timely handling of high-risk issues like fraud or failed transactions.
- **Named Entity Recognition (NER)**
  The system extracts important entities such as account numbers, transaction IDs, dates, and amounts to enhance contextual understanding and improve response accuracy.
- **Context-Aware Response Generation**
  By leveraging Large Language Models (LLMs), the system generates accurate, relevant, and human-like responses tailored to the user's query.
- **Multi-Channel Support**
  The solution supports multiple input sources such as email, chat, and APIs, enabling seamless integration with existing customer communication platforms.
- **Workflow Automation and Routing**
  The system intelligently routes queries to the appropriate department or triggers escalation workflows when human intervention is required.
- **Scalability and Flexibility**
  Designed to handle a large volume of queries efficiently, the system can scale as the number of users or requests increases.
- **Reduced Response Time**
  Automation significantly decreases the time required to process and respond to

customer queries.

- **Improved Accuracy and Consistency**
  AI-based processing ensures consistent and accurate handling of similar queries, reducing human errors.

- **Logging and Monitoring**
  All interactions are recorded for tracking, analysis, and future improvements in system performance

Overall Architecture / Workflow

The Finance Support Triage Agent follows a modular and layered architecture designed to efficiently process, analyze, and respond to customer queries. The system integrates multiple AI components and workflow modules to ensure accurate and timely handling of requests.**User Input Layer**

The overall workflow of the system consists of the following stages:

1. **Input Layer (Data Ingestion)**
   Customer queries are received through multiple channels such as email, chat interfaces, or APIs. This layer ensures that all inputs are collected and forwarded to the processing system.
2. **Preprocessing Layer**
   The input data is cleaned and standardized by removing noise, correcting formatting issues, and preparing the text for analysis.
3. **Classification Engine (Intent + Urgency Detection)**
   The system analyzes the query to determine:
   - The **intent** (e.g., transaction issue, account problem)
   - The **urgency level** (high, medium, low)
     This helps in proper categorization and prioritization.
4. **Named Entity Recognition (NER) Layer**
   Important entities such as account numbers, transaction IDs, dates, and amounts are extracted to understand the context of the query.Data Storage (Optional)

5. Context Builder

This module combines extracted entities with historical or session data to create a complete and meaningful representation of the query.

6. LLM Response Generator

A Large Language Model generates context-aware and human-like responses based on the processed information.

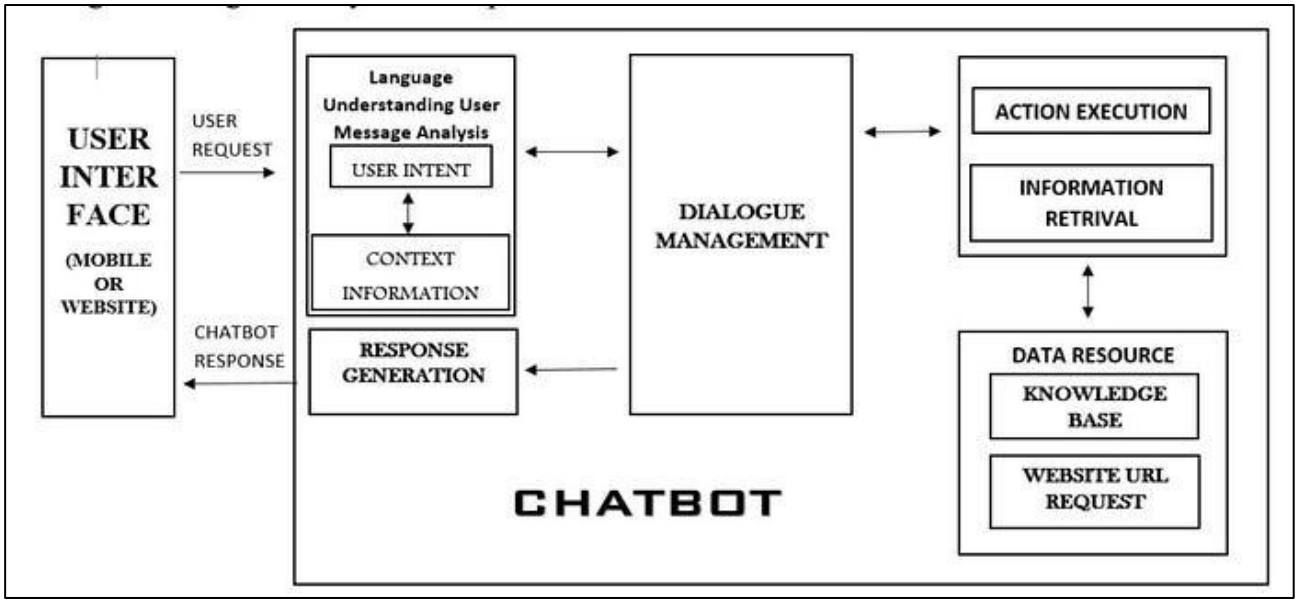
7. Workflow Orchestrator

This component decides the next action:

- Send automated response
- Route to the appropriate department
- Escalate to a human agent

8. Output Layer

The final response is delivered through the same channel (email/chat/API), and the interaction is logged for monitoring and future improvements.



Tools s Technologies Used

1. Input Layer

- User sends query via:
  - Email / Chat / API
- Example inputs:
  - Transaction issue
  - Account problem
  - Fraud complaint

2. Frontend Interface

- Technologies: HTML, CSS, JavaScript
- Features:
  - Chat UI / Form input
  - Displays responses
  - Sends request to backend

3. Backend API

- Technologies: Java (Servlet/JSP) or Python (FastAPI)
- Responsibilities:
  - Receive request
  - Call processing modules
  - Handle business logic

4. AI Processing (Core System)

- **Preprocessing Module** → Cleans input
- **Classification Engine** → Detects intent & urgency
- **NER Module** → Extracts entities
- **LLM Model** → Generates response

5. Storage Layer

- Database (MySQL / MongoDB)
- Vector DB (optional for AI context)
- Stores:
  - Queries
  - Responses
  - Logs

6. Output Layer

- System returns:
  - Automated reply
  - Escalation to agent
  - Ticket generation

1. **Results s Output**

The implementation of the Finance Support Triage Agent demonstrates significant improvements in the automation and efficiency of handling customer support queries in the financial domain. The system successfully processes user inputs, classifies queries, extracts relevant information, and generates appropriate responses.

1.1 System Results

- The system is able to **accurately classify customer queries** into categories such as transaction issues, account problems, fraud alerts, and general inquiries.
- It effectively **identifies the urgency level** of queries, ensuring that critical issues are prioritized and handled promptly.
- Using Named Entity Recognition (NER), the system successfully **extracts key information** such as transaction IDs, account numbers, dates, and amounts.

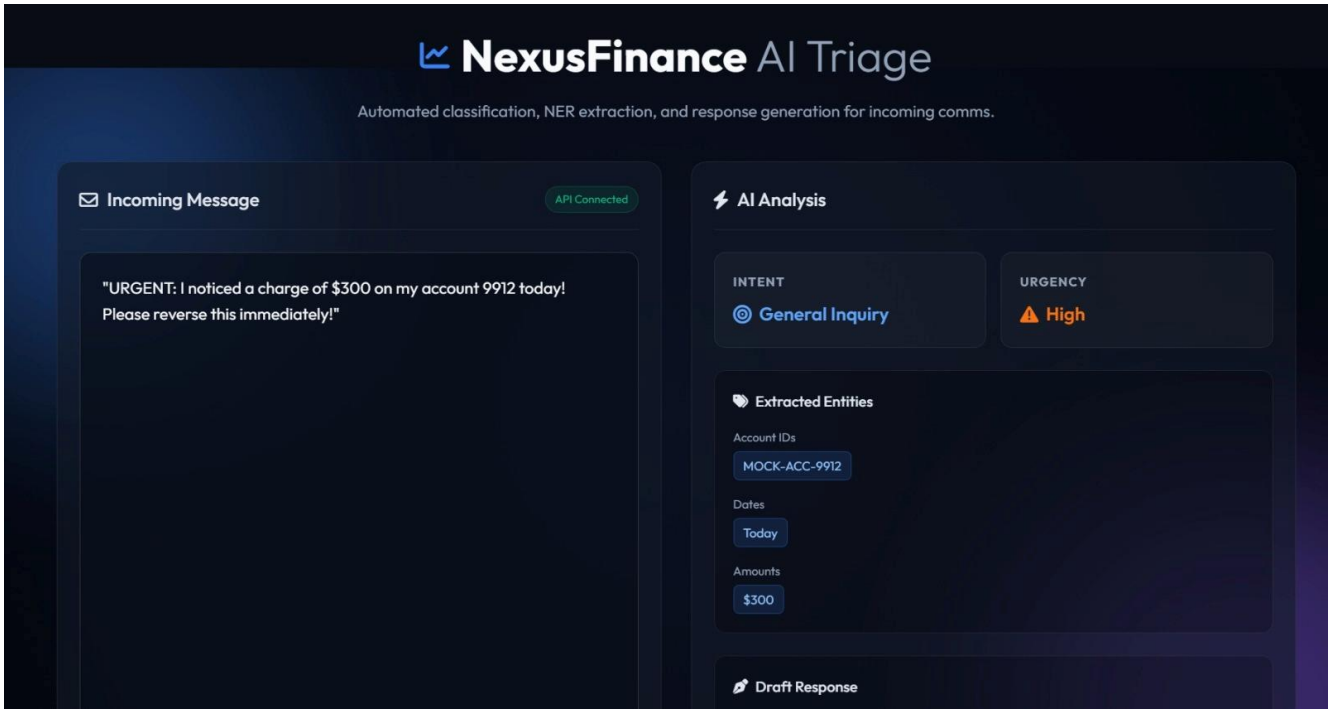
- The integration of Large Language Models (LLMs) enables the system to **generate context-aware and meaningful responses**.
- The workflow automation ensures that queries are **either resolved automatically or escalated** to the appropriate department when necessary.

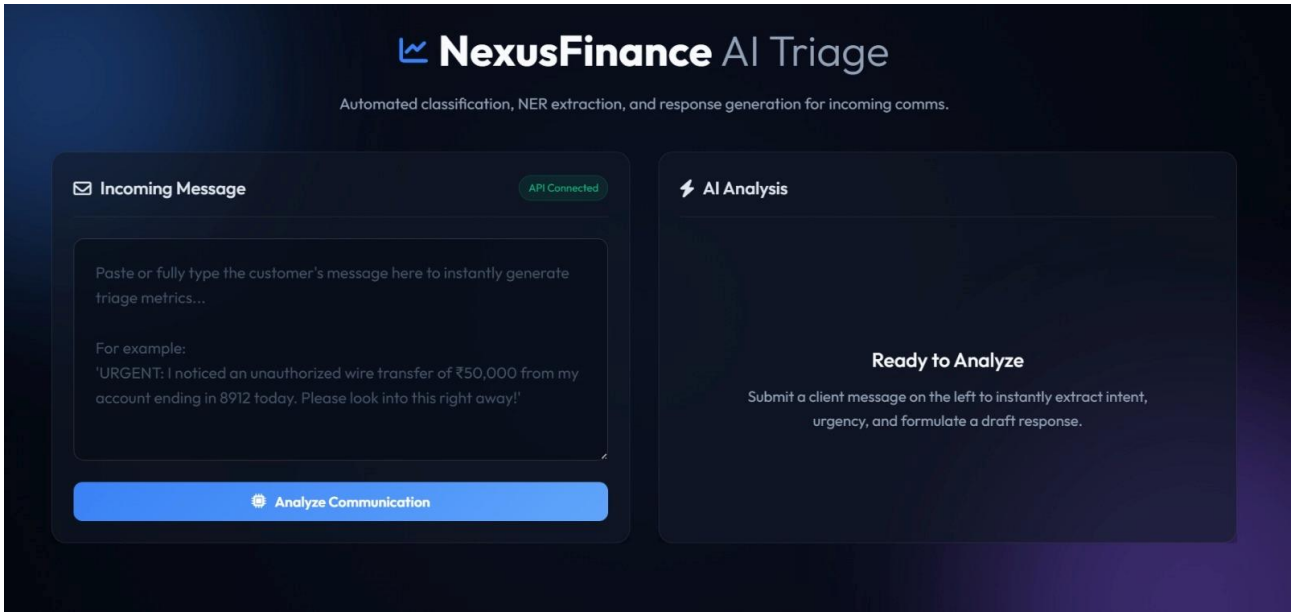
1.2 Output of the System

The system produces the following types of outputs:

- **Automated Responses**
  Instant replies are generated for common queries, reducing response time and manual effort.
- **Escalation Notifications**
  High-priority or complex queries are forwarded to human agents for further handling.
- **Ticket Generation**
  Support tickets are created for tracking and managing unresolved issues.
- **Structured Data Logs**
  All interactions are stored in the database for monitoring, analysis, and future improvements.

System Outputs (Screenshots / Examples)




NexusFinance AI Triage
Automated classification, NER extraction, and response generation for incoming comms.
Incoming Message
API Connected
Paste or fully type the customer's message here to instantly generate triage metrics...
For example:
'URGENT: I noticed an unauthorized wire transfer of ₹50,000 from my account ending in 8912 today. Please look into this right away!'
Analyze Communication
AI Analysis
Ready to Analyze
Submit a client message on the left to instantly extract intent, urgency, and formulate a draft response.

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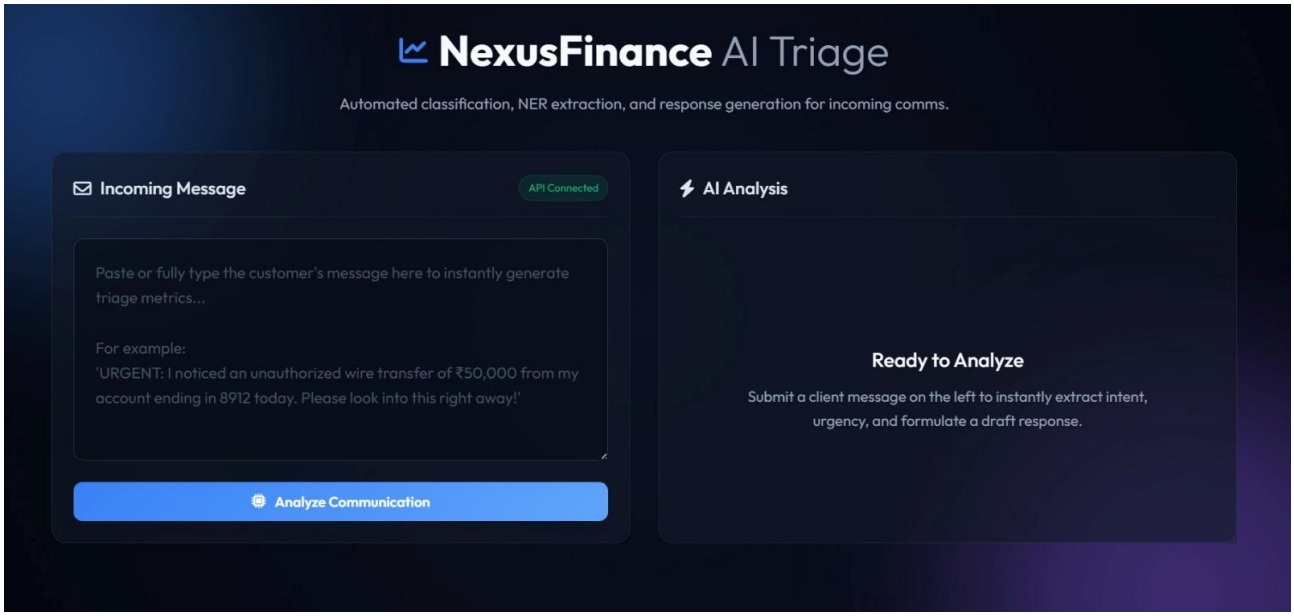
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AI Analysis

INTENT

General Inquiry

URGENCY

High

Extracted Entities

Account IDs

MOCK-ACC-9912

Dates

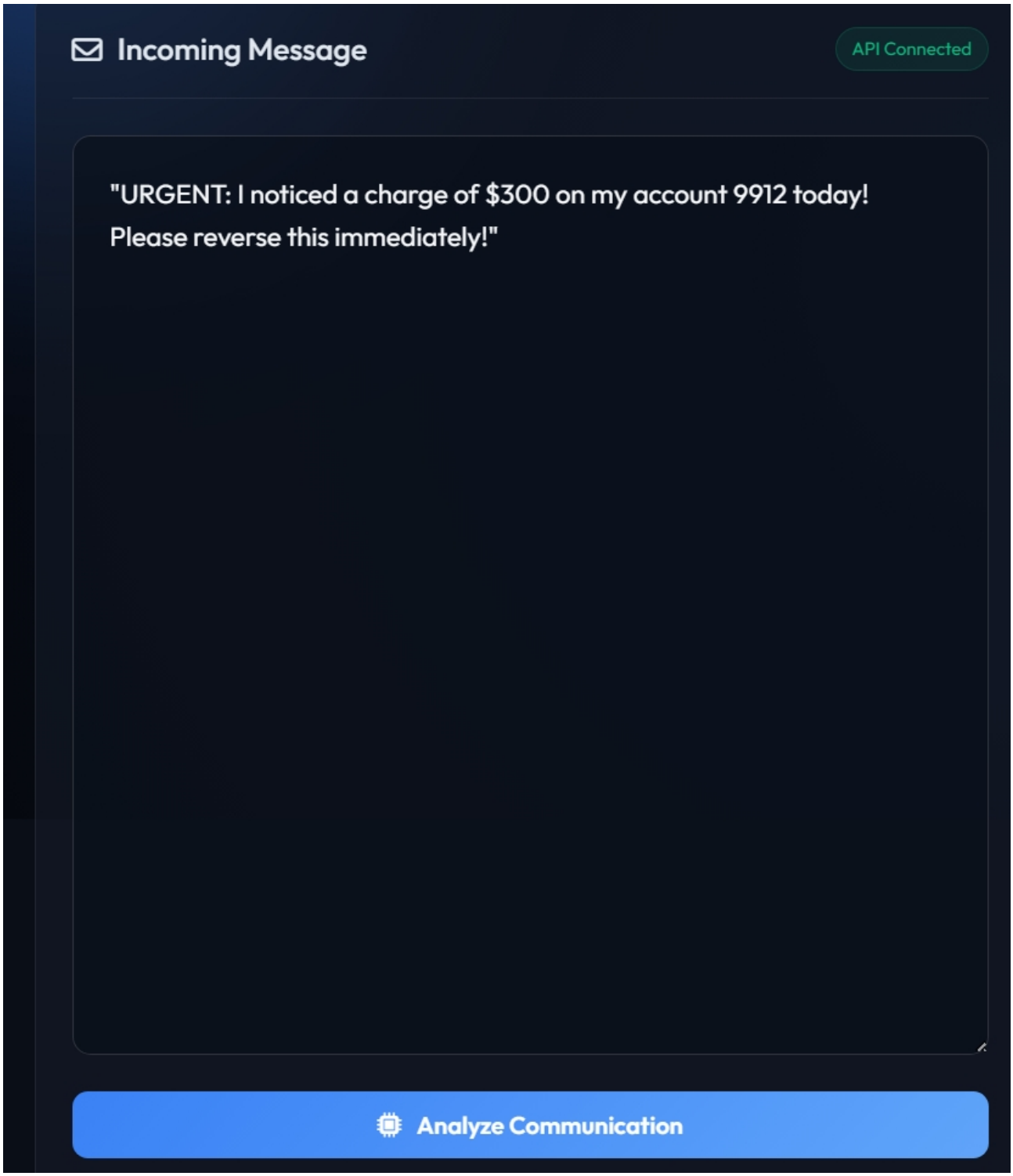
Today

Amounts

$300

Draft Response

Thank you for reaching out regarding your general inquiry. We have prioritized your message as HIGH urgency. Our specialized finance team is currently reviewing your request and details. We will get back to you shortly with next steps.


Incoming Message
API Connected
"URGENT: I noticed a charge of $300 on my account 9912 today! Please reverse this immediately!"
Analyze Communication

AI Analysis

INTENT

General Inquiry

URGENCY

High

Extracted Entities

Account IDs

MOCK-ACC-9912

Dates

Today

Amounts

$300

Draft Response

Thank you for reaching out regarding your general inquiry. We have prioritized your message as HIGH urgency. Our specialized finance team is currently reviewing your request and details. We will get back to you shortly with next steps.

1. Key Outcomes

The development and implementation of the Finance Support Triage Agent resulted in several measurable and qualitative improvements in handling financial customer support queries. The key outcomes of the project are as follows:

- **Efficient Query Handling**
  The system successfully automates the processing of customer queries, reducing dependency on manual intervention.
- **Improved Classification Accuracy**
  Queries are accurately categorized based on intent (transaction, account, fraud, etc.), ensuring proper routing and handling.
- **Effective Prioritization of Critical Issues**
  Urgency detection enables faster handling of high-priority cases such as fraud alerts and failed transactions.
- **Enhanced Response Quality**
  Context-aware responses generated using AI models improve the relevance and clarity of replies.
- **Reduced Response Time**
  Automation significantly decreases the time required to respond to customer queries.
- **Optimized Resource Utilization**
  Human agents can focus on complex issues, while routine queries are handled automatically.
- **Scalable System Design**
  The architecture supports handling large volumes of queries without performance degradation.
- **Consistent and Reliable Support**
  Standardized processing ensures uniform responses and reduces human error.
- **Data-Driven Insights**
  Stored logs and interactions can be analyzed for improving system performance and decision-making.

- **Improved Customer Satisfaction**
  Faster and more accurate responses lead to a better overall user experience.

Conclusion

The Finance Support Triage Agent successfully demonstrates the application of Artificial Intelligence in automating customer support operations within the financial domain. The system effectively integrates Natural Language Processing (NLP), Named Entity Recognition (NER), and Large Language Models (LLMs) to analyze, classify, and respond to customer queries in an efficient and intelligent manner.

By automating key processes such as query classification, urgency detection, entity extraction, and response generation, the system significantly reduces response time and minimizes manual effort. It ensures that critical issues are identified and prioritized appropriately, thereby improving the overall quality and reliability of support services.

The modular architecture of the system allows for scalability and flexibility, making it adaptable to increasing volumes of queries and integration with various communication channels. Additionally, the ability to generate context-aware responses enhances customer satisfaction by providing accurate and timely assistance.

In conclusion, the proposed solution addresses the limitations of traditional support systems and provides a smart, efficient, and scalable approach to managing financial customer queries. The project highlights the potential of AI-driven systems in transforming customer support services and sets a foundation for further advancements in intelligent automation.

Future Scope s Enhancements

The current implementation of the Finance Support Triage Agent provides a strong foundation for automating customer support in the financial domain. However, there are several opportunities for further improvement and expansion to enhance its capabilities and real-world applicability.

Future Scope

- The system can be extended to support **real-time deployment in banking and financial institutions**, enabling live customer interaction at scale.
- Integration with **core banking systems and payment gateways** can allow the system to provide more accurate and real-time updates regarding transactions and account details.
- The solution can be enhanced to support **multi-language processing**, making it accessible to a wider range of users across different regions.
- Advanced **fraud detection mechanisms** can be incorporated using machine learning models to identify suspicious activities more effectively.
- The system can be expanded to include **voice-based support** using speech recognition and text-to-speech technologies.

Enhancements

- Implementation of **personalization features**, where responses are tailored based on user history and preferences.
- Use of **advanced analytics and dashboards** to monitor system performance, query trends, and customer behavior.
- Integration with **mobile applications** to provide seamless support on smartphones.
- Improvement in **model accuracy and training** by using domain-specific datasets and continuous learning techniques.
- Addition of **feedback mechanisms** to collect user input and improve system responses over time.
- Strengthening **security and data privacy measures** to ensure compliance with financial regulations.
- Incorporation of **explainable AI (XAI)** techniques to make the system's decisions more transparent and trustworthy.

Abstract

In the financial services sector, handling a large volume of customer support queries efficiently and accurately is a critical challenge. Traditional systems often rely on manual processes or basic automation, leading to delayed responses, inconsistent query handling, and increased operational costs. To address these issues, this project presents an intelligent **Finance Support Triage Agent** that leverages Artificial Intelligence to automate and optimize customer support operations.

The proposed system utilizes advanced technologies such as Natural Language Processing (NLP), Named Entity Recognition (NER), and Large Language Models (LLMs) to understand and process user queries. It is capable of classifying queries based on intent, detecting urgency levels, extracting relevant entities such as transaction details, and generating context-aware responses.

The architecture of the system follows a modular approach, consisting of input handling, preprocessing, classification, entity extraction, context building, response generation, and output delivery. This ensures scalability, flexibility, and efficient integration with multiple communication channels such as email, chat, and APIs.

The implementation of the Finance Support Triage Agent results in reduced response time, improved accuracy, optimized resource utilization, and enhanced customer satisfaction. The system demonstrates the potential of AI-driven solutions in transforming customer support services within the financial domain and provides a foundation for future enhancements such as real-time deployment, multilingual support, and advanced analytics.

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